

National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	Differential Equations	Course Code:	MT-1006
	Program:	BEE-2	Semester:	Spring 2023
	Duration:	60 minutes	Total Marks:	30
	Paper Date:	08-04-23	Weight	15%
	Section:	All	Page(s):	1
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Student Name:				

Question #1: [CLO-2, 10 marks].

(a) Solve the given second order Cauchy-Euler equation.

$$x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = 0$$

(b) Find the general solution by using Variation of Parameters method.

$$y'' - 4y' + 4y = \sinh x e^{2x}$$

Question #2: [CLO-3, 10 marks]

Find the current in an LRC series circuit when $L = \frac{1}{2}$ henry, $R = 10$ ohms,

$C = 0.01$ farad, $E(t) = 150$ V, $q(0) = 1$ coulombs, and $i(0) = 0$. What is the charge on the capacitor after a long time?

Question #3: [CLO-5, 10 marks]

Find the two infinite power series solutions of the following differential equation

$$y'' + \ln(1-x)y = 0$$

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Question #1: [CLO-01, 10 marks]

- ✓(a) State the order of the following equation and determine whether the following equation is linear or nonlinear.

Second order

$$\left(\frac{d^2x}{dt^2}\right)^3 + 1 = 0. \text{ Non-Linear.}$$

- (b) Verify that the indicated function is an explicit solution of the given differential equation and determines the largest interval of validity.

$$\frac{dy}{dx} = -\frac{1}{2}y^3(1 + \tan x); \quad y = (1 - \tan x)^{-\frac{1}{2}}$$

Question #2: [CLO-02, 20 marks].

Solve the given DE and find its solution.

$$(2x^2 + 3y^2 - 20) \frac{dy}{dx} = -xy$$

Question #3: [CLO-03, 10 marks].

The population of a town grows at a rate proportional to the population present at time t . The initial population of 649 increases by 13% in 6.5 years. What will be the population in 27.5 years? How fast is the population growing at $t = 30$?